

School of Computer Science

A Level 3 Module COMP3056 Autumn 2024/25

Professional Ethics in Computing

Instructions:

Time allowed: Two Hours

Answer ALL questions

Candidates may complete the front cover of their answer book and sign their desk card. **DO NOT turn the examination paper over until instructed to do so.**

Permitted resources:

Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination.

No calculators are permitted in this examination.

Prohibited resources:

All other dictionaries, including subject specific translation dictionaries and electronic dictionaries. Electronic devices capable of storing and retrieving text.

Appended material: NONE

Additional material: British Computer Society Code of Conduct

Information for Invigilators: open book exam

Exam Papers must not be removed from the examination venue

1. Answer **all** parts of this question

(a) This is a question about *Professionalism and Computing*.

i) Give at least 5 characteristics of professionalism according to Bayles. lec4-p11

[5 marks]

1. Provides an important service to society
2. Requires extensive training
3. Involves significant intellectual effort
4. Organisation of members
5. Individual autonomy
6. Certification or licensing

ii) Give reasons why Computing may or may not be considered a profession.

Lack of accreditation, certification or licensing:

[3 marks] lec4-p13

No single organisation of members for the computing profession

But: The harms enabled by computing mean that people doing computing as a job of work still have important ethical obligations

(b) This is a question about *EDI*.

"Programmers need ethics when designing the technologies that influence people's lives."

i) Describe what is meant by equity, diversity, and inclusion. lec6-p2

Equity: Give everyone an even opportunity to participate and achieve.

[4 marks]

Diversity: Encourage and celebrate differences between people.

Inclusion: Everyone is included and belongs.

ii) What actions can be taken to promote EDI in the workplace? lec6-p11

[3 marks]

1. Awareness / vigilance.
2. Remedial action (e.g. improving conditions for disabled)
3. Positive action (e.g. encouraging diversity)
4. Self-reflection on unconscious bias.
5. UNNC strategy and policy on EDI

iii) Suppose you develop an online calculator. Briefly describe two features

that could be introduced to improve inclusion.

[2 marks]

1. The system could provide adjustable font sizes to support elderly users.
2. It could also offer voice input and output to assist users with physical disabilities or visual impairments.
3. In addition, a colour-blind mode would improve accessibility for users with colour vision deficiencies.

(c) This is a question about *Trustworthy Software and Computer Failure*.

Consider this scenario:

A software engineer Pete and his team design and develop a system that monitors the condition of a suite of 100 electric vehicles used in a warehouse. The software reads sensor information and generates an alert if there is a mechanical failure with a vehicle. The alert is sent electronically to one or more human mechanics who are available to fix the problem.

Since the vehicles share the warehouse space with human workers, Pete is aware that this is a safety critical system and arranges a series of rigorous tests with his colleagues, including an independent review. Overall, the system runs through 6 rounds of testing and bug fixes until no further bugs are found. Pete's team also provides detailed documentation and offers extensive training for the mechanics.

Unfortunately, two months after software release one of the vehicles crashes into a box causing damage and a small fire. An investigation shows that at the time of the crash, the system generated an alert about the braking system on this vehicle. There were 3 human mechanics on duty but two of them had overridden the system and so did not spot the alert. The third mechanic saw the alert, but had not attended the training – due to lack of time – and so was not clear what follow-up action to take, if any.

- i) Apply Birsch's criteria for moral responsibility to determine if Pete was responsible for the accident. State your reasoning clearly. lec9-p19
[5 marks]

- ii) In response to this incident, propose up to three further software features that could be added to improve the safety of this system.

i) 3 criteria must be met for a person to bear moral responsibility: [3 marks]

1. The person's actions must have caused the harm, or been a significant causal factor
2. The person must have intended or willed the harm, or it must be a result of his/her negligence, carelessness, or recklessness
3. The person must be able to have known, or must know the consequences of the action, or must have deliberately remained ignorant of them

According to Birsch's criteria, Pete should not be regarded as the primary morally responsible agent. First, in terms of causation, the system successfully generated the alert, indicating that the software itself functioned as intended. The immediate cause of the accident was the mechanics' decision to override the alert and the lack of appropriate follow-up action by an untrained operator.

Second, regarding foreseeability, although human misuse of alerts is generally foreseeable, Pete took reasonable and proportionate steps to mitigate such risks, including extensive testing, independent review, detailed documentation, and training provision.

Finally, there is no clear evidence of negligence on Pete's part. The failure arose primarily from human misuse and organisational shortcomings in enforcing training and operational procedures.

Therefore, moral responsibility should be understood as distributed among the mechanics and the organisation's management rather than resting with Pete alone.

ii)

The system could be improved by introducing stricter override controls, requiring mechanics to explicitly confirm overrides, record reasons, and limit the duration for which an alert can be suppressed.

In addition, critical alerts should be designed with redundancy, such as repeated notifications and multi-modal warnings (e.g. visual and audio alerts), ensuring that alerts cannot be ignored unintentionally.

The system could also provide step-by-step guidance on required follow-up actions and prevent alerts from being dismissed until corrective actions are confirmed.

Finally, access to the system should be restricted so that only trained and authorised mechanics are able to log in and respond to safety-critical alerts.

2. Consider this scenario:

Software development company WombatAI is building an AI-powered computer system called CIP to price car insurance for customers of a big insurance company. In tests, the new CIP system works well and simulations demonstrate that using it will save the insurance company money by more efficiently pricing drivers based on characteristics such as age, job details, previous driving experience, and demographics such as location. Data about ethnicity and gender are also collected for testing purposes but are not used in training. Although the system works well, no one fully understands why since it uses a deep neural network as part of its decision process.

During testing the development team discover:

- [1] For a small group of young drivers, there is a disproportionate increase in price. This group of drivers makes up less than 1% of the population, and further analysis shows many of these young drivers are also from an ethnic minority.
- [2] Insurance price is, on average, the same for people in different geographical regions, but it is noticed that people in one urban location, L, are less risky than those in other regions. This region represents 20% of the population.

A meeting is convened to discuss these issues and whether they represent discrimination. The manager of the software development team asks all developers to contribute to the meeting and together they form the following views.

- For the first point, the development team decides that the effect is only noticeable within a very small group and this is within the projects “bounds of error”.
- For the second, some developers say this is unfair for people living in L since they are paying a higher premium than they should based on their risk profile. However, others point out that the insurance company wants parity in pricing between regions so that there is fairness between regions.

Therefore, they decide that no action is required to follow up on either of these points.

Answer these questions about the ethics and fairness of the decisions made by the software developers.

- (a) Consider this scenario from the perspective of four different Theories of Ethics; i.e. categorical imperative, utilitarianism, contractarianism, ethics of care. Evaluate the decisions made in the scenario by applying each Theory of Ethics. What does this analysis tell us about whether the decisions are right or wrong? [10 marks]
- (b) Consider the problems and decisions in the context of the BCS code of conduct. [6 marks]
- (c) What definition(s) of fairness, if any, are being used in this scenario. Discuss and justify your answer. [6 marks]

(d) What more could the development team do to try to understand how CIP is working and coming to decisions? [3 marks]

a)

1. From a categorical imperative perspective, an action is moral only if it can be stated as a universal rule without contradiction. In this scenario, adopting the rule that unfair impacts on minority groups can be ignored because they fall within an acceptable margin of error cannot be universalised, as it would legitimise systematic discrimination. This also fails to respect individuals as ends in themselves, particularly ethnic minorities who are treated merely as statistical outliers.

2. From a utilitarian perspective, the decision may initially appear justified, as most customers benefit from lower insurance costs. However, the significant disadvantage imposed on younger people from ethnic minorities and residents of Urban L must be included in the overall utility calculation. Because the right decision is the one that causes the most happiness, amongst all people affected by the decision

When long-term social harms such as loss of trust, perceived unfairness, and potential discrimination are considered, the overall happiness may no longer be maximised.

3. From a contractarian perspective, ethical decisions are those that could be accepted by rational members of society under fair conditions. While the public accepts car insurance systems on the assumption that pricing rules are reasonable and justifiable, ignoring systematic disadvantages to certain groups undermines this social agreement.

Therefore, failing to address these issues violates the common interests underpinning the social contract.

4. From the ethics of care perspective, moral action requires attentiveness to the needs of those who may be vulnerable or dependent. The higher insurance prices imposed on younger people from ethnic minorities and residents of Urban L demonstrate a lack of care for those groups.

Treating these impacts as acceptable errors reflects indifference rather than responsiveness to concrete harm.

Taken together, these ethical perspectives suggest that the team's decision is ethically problematic.

Rather than dismissing the issues as acceptable errors, the development team should conduct further investigation, improve transparency, and reassess the system before deployment.

b)

Public Interest (BCS 1a, 1c, 1d):

By ignoring systematic disadvantages faced by ethnic minorities and residents of Urban City L, the development team fails to have due regard for the wellbeing of others and risks discriminatory outcomes. This also undermines the promotion of equal access to the benefits of IT.

Professional Competence and Integrity (BCS 2a, 2c, 2f):

Deploying a black-box insurance pricing system without adequately understanding why it produces biased outcomes demonstrates a lack of professional competence and due diligence. This negligent inaction foreseeably causes harm to affected user groups.

c)

In computer science, fairness is often formalised using statistical definitions such as independence, separation, and sufficiency.

In this scenario, the insurance pricing system fails to satisfy independence, as pricing outcomes are correlated with sensitive attributes such as ethnicity and age, with younger people from ethnic minorities being charged higher premiums.

There is insufficient information to determine whether the system satisfies separation, since the scenario does not disclose whether error rates are equal across groups given the same underlying risk.

Furthermore, the system violates sufficiency, as identical or higher pricing outcomes correspond to different true risk levels, particularly for residents of Urban City L whose actual risk is lower.

This demonstrates that the system adopts a narrow, outcome-based notion of fairness while failing to ensure fairness across different groups.

d)

The team could apply Explainable AI (XAI) techniques such as LIME, SHAP, or PDP(partial dependence plots) to understand which features are driving the pricing decisions, particularly for ethnic minorities and residents of Urban City L.

In addition, they could perform subgroup analysis to examine whether different demographic groups are systematically disadvantaged.

Finally, reviewing the training data and model design would help identify potential data bias or proxy variables influencing the system's decisions.

End